HW4

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Q1

Let $\underline{x}$ be an $n \times 1$ random vector with mean vector $\underline{\mu}$ and variance-covariance matrix Σ. Use Theorem 3.1 to show that $\underline{x}$ has a multivariate normal distribution if and only if, for every $n \times 1$ vector of constants $\underline{a}$, $\underline{a}'\underline{x}$ has a (univariate) normal distribution.

Answer

Note: As in the definition of the multivariate normal, we take $\operatorname{rank}(\Sigma) = k \geq 1$, and we read $N(c, 0)$ as the point mass at c (needed when $\underline{a}'\Sigma\underline{a} = 0$).

We utilize the following results:

- Theorem 3.1 (uniqueness of mgfs): if the mgfs of two random vectors exist and are equal on an open rectangle containing the origin, then their cdfs are identical.
- Definition (n-variate normal of rank k): $\underline{x} \sim N(\underline{\mu}, \Sigma)$ if $\underline{x} \overset{d}{=} \underline{\mu} + \Gamma'\underline{z}$, where Γ is $k \times n$ with $\Sigma = \Gamma'\Gamma$ (such a Γ exists by Lemma 2.15) and $\underline{z} \sim N(\underline{0}_k, I_k)$.
- mgf of the multivariate normal (Section 3.1): if $\underline{x} \sim N(\underline{\mu}, \Sigma)$, then $M_{\underline{x}}(\underline{t}) = \exp\left(\underline{t}'\underline{\mu} + \frac{1}{2}\underline{t}'\Sigma\underline{t}\right)$ for all $\underline{t} \in \mathbb{R}^n$; in particular, $N(m, v)$ has mgf $\exp\left(tm + \frac{1}{2}t^2 v\right)$, $t \in \mathbb{R}$.
- Corollary to Lemma 2.4: $\operatorname{Var}(L'\underline{x}) = L'\Sigma L$, where $\Sigma = \operatorname{Var}(\underline{x})$.

Proof: For any $\underline{a} \in \mathbb{R}^n$ and $t \in \mathbb{R}$,

$$M_{\underline{a}'\underline{x}}(t) = \mathrm{E}\left[\exp(t\underline{a}'\underline{x})\right] = \mathrm{E}\left[\exp\left((t\underline{a})'\underline{x}\right)\right] = M_{\underline{x}}(t\underline{a}), \tag{Q1}$$

whenever either side exists.

($\Rightarrow$) Suppose $\underline{x}$ is multivariate normal, so that $\underline{x} \sim N(\underline{\mu}, \Sigma)$, and let $\underline{a} \in \mathbb{R}^n$ be given. By (Q1) and the mgf of $\underline{x}$, for every $t \in \mathbb{R}$,

$$M_{\underline{a}'\underline{x}}(t) = \exp\left(t\underline{a}'\underline{\mu} + \frac{1}{2}t^2\underline{a}'\Sigma\underline{a}\right),$$

which is the mgf of $N(\underline{a}'\underline{\mu}, \underline{a}'\Sigma\underline{a})$. By Theorem 3.1, $\underline{a}'\underline{x} \sim N(\underline{a}'\underline{\mu}, \underline{a}'\Sigma\underline{a})$. Since $\underline{a}$ was arbitrary, every $\underline{a}'\underline{x}$ is (univariate) normal.

($\Leftarrow$) Suppose $\underline{a}'\underline{x}$ is normal for every $\underline{a} \in \mathbb{R}^n$, and let $\underline{t} \in \mathbb{R}^n$ be given. Since $\mathrm{E}(\underline{t}'\underline{x}) = \underline{t}'\underline{\mu}$ and $\operatorname{Var}(\underline{t}'\underline{x}) = \underline{t}'\Sigma\underline{t}$ (Corollary to Lemma 2.4), $\underline{t}'\underline{x} \sim N(\underline{t}'\underline{\mu}, \underline{t}'\Sigma\underline{t})$, so (Q1) with $t = 1$ gives

$$M_{\underline{x}}(\underline{t}) = M_{\underline{t}'\underline{x}}(1) = \exp\left(\underline{t}'\underline{\mu} + \frac{1}{2}\underline{t}'\Sigma\underline{t}\right).$$

Since $\underline{t}$ was arbitrary, $M_{\underline{x}}$ exists on all of $\mathbb{R}^n$. Now take Γ $(k \times n)$ with $\Sigma = \Gamma'\Gamma$ and $\underline{z} \sim N(\underline{0}_k, I_k)$. By definition, $\underline{\mu} + \Gamma'\underline{z} \sim N(\underline{\mu}, \Sigma)$, so its mgf is also $\exp\left(\underline{t}'\underline{\mu} + \frac{1}{2}\underline{t}'\Sigma\underline{t}\right)$ for all $\underline{t} \in \mathbb{R}^n$. The two mgfs agree on $\mathbb{R}^n$, an open rectangle containing the origin, so by Theorem 3.1, $\underline{x} \overset{d}{=} \underline{\mu} + \Gamma'\underline{z}$; that is, $\underline{x} \sim N(\underline{\mu}, \Sigma)$.

Hence $\underline{x}$ is multivariate normal if and only if every linear combination $\underline{a}'\underline{x}$ is univariate normal.

Q2

Let $\underline{x} \sim N(\underline{\mu}, \Sigma)$. Show that

$$\operatorname{cov}(\underline{x}'A\underline{x}, \underline{x}'B\underline{x}) = 2\operatorname{tr}(A\Sigma B\Sigma) + 4\underline{\mu}'A\Sigma B\underline{\mu},$$

where A and B are symmetric matrices of constants. [Hint: For any random variables w and z, $\operatorname{Var}(w+z) = \operatorname{Var}(w) + \operatorname{Var}(z) + 2\operatorname{cov}(w,z)$.]

Answer

We utilize the following results:

- Variance of a quadratic form (Section 3.2): if $\underline{x} \sim N(\underline{\mu}, \Sigma)$ and C is a symmetric matrix of constants, then $\operatorname{Var}(\underline{x}'C\underline{x}) = 2\operatorname{tr}\left[(C\Sigma)^2\right] + 4\underline{\mu}'C\Sigma C\underline{\mu}$.

Proof: Following the hint, write $w \equiv \underline{x}'A\underline{x}$ and $z \equiv \underline{x}'B\underline{x}$, so that $w+z = \underline{x}'(A+B)\underline{x}$ with $A+B$ symmetric. The Section 3.2 result with $C = A + B$ gives

$$\operatorname{Var}(w+z) = 2\operatorname{tr}\left[(A\Sigma + B\Sigma)^2\right] + 4\underline{\mu}'(A+B)\Sigma(A+B)\underline{\mu}.$$

Expanding each term,

$$\begin{aligned}
\operatorname{tr}\left[(A\Sigma + B\Sigma)^2\right] &= \operatorname{tr}\left[(A\Sigma)^2\right] + \operatorname{tr}\left[(B\Sigma)^2\right] + \operatorname{tr}(A\Sigma B\Sigma) + \operatorname{tr}(B\Sigma A\Sigma) \\
&= \operatorname{tr}\left[(A\Sigma)^2\right] + \operatorname{tr}\left[(B\Sigma)^2\right] + 2\operatorname{tr}(A\Sigma B\Sigma), \\
\underline{\mu}'(A+B)\Sigma(A+B)\underline{\mu} &= \underline{\mu}'A\Sigma A\underline{\mu} + \underline{\mu}'B\Sigma B\underline{\mu} + \underline{\mu}'A\Sigma B\underline{\mu} + \underline{\mu}'B\Sigma A\underline{\mu} \\
&= \underline{\mu}'A\Sigma A\underline{\mu} + \underline{\mu}'B\Sigma B\underline{\mu} + 2\underline{\mu}'A\Sigma B\underline{\mu},
\end{aligned}$$

since $\operatorname{tr}(B\Sigma A\Sigma) = \operatorname{tr}(A\Sigma B\Sigma)$ (cyclic property of the trace), and the scalar $\underline{\mu}'B\Sigma A\underline{\mu}$ equals its transpose $\underline{\mu}'A\Sigma B\underline{\mu}$ (A, B, and Σ are symmetric). Applying the Section 3.2 result with $C = A$ and $C = B$ as well, it then follows that

$$\operatorname{Var}(w+z) = \operatorname{Var}(w) + \operatorname{Var}(z) + 4\operatorname{tr}(A\Sigma B\Sigma) + 8\underline{\mu}'A\Sigma B\underline{\mu}.$$

Comparing with the hint, $2\operatorname{cov}(w,z) = 4\operatorname{tr}(A\Sigma B\Sigma) + 8\underline{\mu}'A\Sigma B\underline{\mu}$; that is,

$$\operatorname{cov}(\underline{x}'A\underline{x}, \underline{x}'B\underline{x}) = 2\operatorname{tr}(A\Sigma B\Sigma) + 4\underline{\mu}'A\Sigma B\underline{\mu}. \tag{Q2}$$

Q3

Let $\underline{x} \sim N(\underline{\mu}, \Sigma)$, where $\underline{x}$ is an $n \times 1$ vector and $\text{rank}(\Sigma) \geq 1$. Let

$$w_1 = \underline{b}_1'\underline{x} + \underline{x}'A_1\underline{x} \quad \text{and} \quad w_2 = \underline{b}_2'\underline{x} + \underline{x}'A_2\underline{x},$$

where $\underline{b}_1$ and $\underline{b}_2$ are vectors of constants and A_1 and A_2 are symmetric matrices of constants. Let

$$\underline{c}_1 = \underline{b}_1 + 2A_1\underline{\mu} \quad \text{and} \quad \underline{c}_2 = \underline{b}_2 + 2A_2\underline{\mu}.$$

Note: Each $\underline{b}_i$ is $n \times 1$ and each A_i is $n \times n$, so each $\begin{pmatrix} \underline{b}_i' \\ A_i \end{pmatrix}$ is $(n+1) \times n$. (The vectors $\underline{c}_1$ and $\underline{c}_2$ do not enter (a) or (b); they are just along for the ride.)

(a)

Find

$$\text{cov}\left(\begin{pmatrix} \underline{b}_1' \\ A_1 \end{pmatrix}\underline{x}, \begin{pmatrix} \underline{b}_2' \\ A_2 \end{pmatrix}\underline{x}\right).$$

Answer

$$\text{cov}\left(\begin{pmatrix} \underline{b}_1' \\ A_1 \end{pmatrix}\underline{x}, \begin{pmatrix} \underline{b}_2' \\ A_2 \end{pmatrix}\underline{x}\right) = \begin{pmatrix} \underline{b}_1'\Sigma\underline{b}_2 & \underline{b}_1'\Sigma A_2 \\ A_1\Sigma\underline{b}_2 & A_1\Sigma A_2 \end{pmatrix}. \qquad \text{(Q3 (a))}$$

We utilize the following results:

- Lemma 2.4: if L and M are matrices of constants with as many rows as $\underline{x}$ has elements, then $\text{cov}(L'\underline{x}, M'\underline{x}) = L'\Sigma M$, where $\Sigma = \text{Var}(\underline{x})$.

Take $L \equiv (\underline{b}_1, A_1)$ and $M \equiv (\underline{b}_2, A_2)$, each $n \times (n+1)$. Since A_1 and A_2 are symmetric, $L' = \begin{pmatrix} \underline{b}_1' \\ A_1 \end{pmatrix}$ and $M' = \begin{pmatrix} \underline{b}_2' \\ A_2 \end{pmatrix}$, so by Lemma 2.4 the covariance is

$$L'\Sigma M = \begin{pmatrix} \underline{b}_1' \\ A_1 \end{pmatrix}\Sigma\begin{pmatrix} \underline{b}_2 & A_2 \end{pmatrix},$$

and multiplying out blockwise gives (Q3 (a)), with blocks of sizes 1×1, $1 \times n$, $n \times 1$, and $n \times n$.

(b)

Suppose that (i) $\underline{b}_1'\Sigma\underline{b}_2 = 0$, (ii) $\underline{b}_1'\Sigma A_2 = \underline{0}'$, (iii) $\underline{b}_2'\Sigma A_1 = \underline{0}'$, and (iv) $A_1\Sigma A_2 = 0$. Use (a) to show that w_1 and w_2 are independent. [Hint: Note that $w_i = \underline{b}_i'\underline{x} + (A_i\underline{x})'A_i^-(A_i\underline{x})$ for $i = 1, 2$.]

Answer

We utilize the following results:

- Corollary to Theorem 3.7: if $\underline{x} \sim N(\underline{\mu}, \Sigma)$ and B_1, B_2 are matrices of constants with n columns, then $B_1\underline{x}$ and $B_2\underline{x}$ are independent if and only if $B_1\Sigma B_2' = 0$ (the corollary's constant vectors play no role here, so we take them to be zero).
- Definition of a generalized inverse: $A_iA_i^-A_i = A_i$.
- Borel functions of independent random vectors are independent.

Proof: For $i = 1, 2$, let $B_i \equiv \begin{pmatrix} \underline{b}_i' \\ A_i \end{pmatrix}$ and $\underline{u}_i \equiv B_i\underline{x} = \begin{pmatrix} \underline{b}_i'\underline{x} \\ A_i\underline{x} \end{pmatrix}$. By Lemma 2.4, $B_1\Sigma B_2' = \text{cov}(\underline{u}_1, \underline{u}_2)$, the matrix in (Q3 (a)), and its four blocks vanish under the assumptions: $\underline{b}_1'\Sigma\underline{b}_2 = 0$ by (i), $\underline{b}_1'\Sigma A_2 = \underline{0}_n'$ by (ii), $A_1\Sigma\underline{b}_2 = (\underline{b}_2'\Sigma A_1)' = \underline{0}_n$ by (iii) (as A_1 and Σ are symmetric), and $A_1\Sigma A_2 = 0$ by (iv). So $B_1\Sigma B_2' = 0$, and by the Corollary to Theorem 3.7, $\underline{u}_1$ and $\underline{u}_2$ are independent.

It then suffices to show that each w_i is a function of $\underline{u}_i$ alone. Since $A_i A_i^- A_i = A_i$ and $A_i' = A_i$,

$$\underline{x}' A_i \underline{x} = \underline{x}' A_i A_i^- A_i \underline{x} = (A_i \underline{x})' A_i^- (A_i \underline{x}),$$

which is the hint. So $w_i = g_i(\underline{b}_i' \underline{x}, A_i \underline{x}) = g_i(\underline{u}_i)$, where $g_i(s, \underline{v}) \equiv s + \underline{v}' A_i^- \underline{v}$ for $s \in \mathbb{R}$ and $\underline{v} \in \mathbb{R}^n$. Since g_1 and g_2 are continuous and $\underline{u}_1$ and $\underline{u}_2$ are independent, $w_1 = g_1(\underline{u}_1)$ and $w_2 = g_2(\underline{u}_2)$ are independent.

Q4

Take the model to be Model II and assume that $\underline{y}$ has a multivariate normal distribution.

Note: Under Model II with $\underline{y}$ multivariate normal, $\underline{y} \sim N(X\underline{\beta}, \sigma^2 I_n)$, where X is $n \times p$ with $p^* \equiv \text{rank}(X)$. We assume $n > p^*$ (otherwise $\underline{y}'(\underline{y} - X\widehat{\underline{\beta}}) = 0$, and every estimator of the given form has MSE σ^4). As in Section 2.8, write $Q(\widehat{\underline{\beta}}) = (\underline{y} - X\widehat{\underline{\beta}})'(\underline{y} - X\widehat{\underline{\beta}}) = \underline{y}'(\underline{y} - X\widehat{\underline{\beta}})$ for the residual sum of squares, and let $\widetilde{\sigma}_k^2 \equiv (1/k)Q(\widehat{\underline{\beta}})$.

We utilize the following results for (a)–(b):

- Corollary to Theorem 2.5 and the Note following Theorem 2.6: for any solution $\widehat{\underline{\beta}}$ to the normal equations, $X\widehat{\underline{\beta}} = P_X\underline{y}$, so $\underline{y} - X\widehat{\underline{\beta}} = (I_n - P_X)\underline{y}$, where $P_X = X(X'X)^- X'$.
- Lemmas 2.6–2.7 and property (4) of P_X: $I_n - P_X$ is symmetric and idempotent with $\text{rank}(I_n - P_X) = n - p^*$, and $P_X X = X$, so $(I_n - P_X)X = 0$.
- Corollary to Lemma 2.5: if F is idempotent, then $\text{tr}(F) = \text{rank}(F)$.
- Lemma 2.8: if $\text{E}(\underline{x}) = \underline{\mu}$ and $\text{Var}(\underline{x}) = \Sigma$, then $\text{E}(\underline{x}'F\underline{x}) = \text{tr}(F\Sigma) + \underline{\mu}'F\underline{\mu}$.
- Variance of a quadratic form (Section 3.2; stated in Q2).

(a)

Obtain the expectation, the variance, and the mean squared error for any estimator of σ^2 having the form $(1/k)\underline{y}'(\underline{y} - X\widehat{\underline{\beta}})$, where k is a nonzero constant, and $\widehat{\underline{\beta}}$ is any solution to the normal equations.

Answer

For every nonzero k,

$$\begin{aligned} \text{E}(\widetilde{\sigma}_k^2) &= \frac{(n-p^*)\sigma^2}{k}, \qquad \text{Var}(\widetilde{\sigma}_k^2) = \frac{2(n-p^*)\sigma^4}{k^2}, \\ \text{MSE}(\widetilde{\sigma}_k^2) &= \frac{\sigma^4}{k^2}\left[2(n-p^*) + (n-p^*-k)^2\right]. \end{aligned} \tag{Q4 (a)}$$

By the first result, $Q(\widehat{\underline{\beta}}) = \underline{y}'(I_n - P_X)\underline{y}$ for any solution $\widehat{\underline{\beta}}$, so $\widetilde{\sigma}_k^2$ does not depend on the choice of $\widehat{\underline{\beta}}$. We apply Lemma 2.8 and the Section 3.2 result with $F = C = I_n - P_X$, $\underline{\mu} = X\underline{\beta}$, and $\Sigma = \sigma^2 I_n$. Since $(I_n - P_X)X\underline{\beta} = \underline{0}_n$, both mean terms vanish, and since $(I_n - P_X)^2 = I_n - P_X$ with $\text{tr}(I_n - P_X) = \text{rank}(I_n - P_X) = n - p^*$,

$$\text{E}[Q(\widehat{\underline{\beta}})] = \sigma^2 \,\text{tr}(I_n - P_X) = (n-p^*)\sigma^2, \qquad \text{Var}[Q(\widehat{\underline{\beta}})] = 2\sigma^4 \,\text{tr}\left[(I_n - P_X)^2\right] = 2(n-p^*)\sigma^4.$$

Dividing by k (resp. k^2) gives the expectation (resp. variance) in (Q4 (a)), and

$$\text{MSE}(\widetilde{\sigma}_k^2) = \text{Var}(\widetilde{\sigma}_k^2) + \left[\text{E}(\widetilde{\sigma}_k^2) - \sigma^2\right]^2 = \frac{2(n-p^*)\sigma^4}{k^2} + \sigma^4\left(\frac{n-p^*}{k} - 1\right)^2,$$

which rearranges to the MSE in (Q4 (a)). (Equivalently, $Q(\widehat{\underline{\beta}})/\sigma^2 \sim \chi^2(n-p^*)$ by Example 1 following Theorem 3.16, with mean $n-p^*$ and variance $2(n-p^*)$.)

(b)

Of those estimators of σ^2 having the above form, which has the smallest mean squared error? (That is, determine the optimum value of k.)

Answer

$k = n - p^* + 2$; that is, $\underline{y}'(\underline{y} - X\widehat{\underline{\beta}})/(n - p^* + 2)$ has the smallest MSE of the family, namely $2\sigma^4/(n - p^* + 2)$.

Proof: Write $m \equiv n - p^* \geq 1$ and $v \equiv 1/k$, which ranges over all nonzero reals as k does. By (Q4 (a)),

$$\frac{\mathrm{MSE}(\widetilde{\sigma}^2_k)}{\sigma^4} = 2mv^2 + (mv-1)^2 = m(m+2)v^2 - 2mv + 1 \equiv h(v).$$

Since $m(m+2) > 0$, h is a strictly convex quadratic, uniquely minimized where $h'(v) = 2m(m+2)v - 2m = 0$, that is, at $v^* = 1/(m+2) \neq 0$. Hence the optimum is $k^* = 1/v^* = m+2 = n - p^* + 2$, with

$$\mathrm{MSE}(\widetilde{\sigma}^2_{k^*}) = \sigma^4 h(v^*) = \sigma^4 \left(1 - \frac{m}{m+2}\right) = \frac{2\sigma^4}{n - p^* + 2}.$$

Said differently, dividing by $n - p^* + 2$ rather than the $n - p^*$ of the unbiased $\widehat{\sigma}^2$ (Section 2.8) accepts a downward bias to cut the MSE from $2\sigma^4/(n-p^*)$ to $2\sigma^4/(n-p^*+2)$.

As a quick numerical check (a one-way layout with an intercept, so X is rank-deficient):

```
set.seed(611)
n <- 12; sigma <- 2
g <- factor(rep(1:3, times = 4))
X <- cbind(1, model.matrix(~ g - 1))            # n x 4, rank 3
pstar <- qr(X)$rank; m <- n - pstar             # Expect pstar = 3, m = 9
beta <- c(1, -1, 0.5, 2)
PX <- X %*% MASS::ginv(crossprod(X)) %*% t(X)
R <- 1e5
Y <- drop(X %*% beta) + matrix(rnorm(n * R, sd = sigma), n, R)
Q <- colSums(Y * ((diag(n) - PX) %*% Y))        # y'(y - X betahat), one per replicate
c(mean(Q), m * sigma^2)                         # Expect approx. equal (36)
```

```
## [1] 35.9919 36.0000
```

```
c(var(Q), 2 * m * sigma^4)                      # Expect approx. equal (288)
```

```
## [1] 287.449 288.000
```

```
mse <- function(k) sigma^4 / k^2 * (2 * m + (m - k)^2)
ks <- m + 0:4
rbind(simulated = sapply(ks, function(k) mean((Q / k - sigma^2)^2)),
      theory = mse(ks))                         # Expect agreement; smallest at k = 11
```

```
##               [,1]    [,2]     [,3]     [,4]     [,5]
## simulated 3.548718 3.03511 2.905585 2.997504 3.217191
## theory    3.555556 3.04000 2.909091 3.000000 3.218935
```

```
optimize(mse, c(1, 50))$minimum                 # Expect m + 2 = 11
```

```
## [1] 10.99999
```

Q5

Let $\underline{x} \sim N(\underline{\mu}, \Sigma)$, where $\underline{x}$ is an $r \times 1$ vector and $\text{rank}(\Sigma) = k \geq 1$ (where possibly $k < r$). Let A be an $r \times r$ symmetric matrix. Show that, if $A\Sigma$ is idempotent of rank s and if in addition $\underline{\mu}$ lies in $\mathcal{C}(\Sigma)$, then $\underline{x}'A\underline{x} \sim \chi^2\left(s, \frac{1}{2}\underline{\mu}'A\underline{\mu}\right)$. (Hint: Let Γ be a $k \times r$ matrix such that $\Sigma = \Gamma'\Gamma$, and show that $A\Sigma$ is idempotent of rank $s \implies \Gamma A\Gamma'$ is idempotent of rank s. Then, use $\underline{x}$ has the same distribution as $\Gamma'(\underline{z} + \underline{a})$, where $\underline{z}$ is a standard normal vector and $\underline{a}$ is a certain vector of constants.)

Answer

Note: Σ is $r \times r$ and Γ is $k \times r$. We take $s \geq 1$, as in Theorem 3.16 (if $s = 0$, the argument below gives $\underline{x}'A\underline{x} = 0$ with probability one).

We utilize the following results:

- Definition of $N(\underline{\mu}, \Sigma)$ (stated in Q1): $\underline{x} \overset{d}{=} \underline{\mu} + \Gamma'\underline{z}$ with $\underline{z} \sim N(\underline{0}_k, I_k)$, for any $k \times r$ matrix Γ with $\Sigma = \Gamma'\Gamma$ (the distribution of $\underline{\mu} + \Gamma'\underline{z}$ depends on Γ only through $\Gamma'\Gamma$, as its mgf shows).
- Lemma 1.3(i): for any matrix X, $PX'X = QX'X$ implies $PX' = QX'$.
- Corollary to Lemma 2.5 (stated in Q4).
- Theorem 3.5: if $\underline{x} \sim N(\underline{\mu}, \Sigma)$, then $B\underline{x} + \underline{b} \sim N(B\underline{\mu} + \underline{b}, B\Sigma B')$.
- Theorem 3.16 (case $\Sigma = I$): if $\underline{v} \sim N(\underline{a}, I_k)$ and M is a $k \times k$ symmetric matrix that is idempotent of rank $s \geq 1$, then $\underline{v}'M\underline{v} \sim \chi^2\left(s, \frac{1}{2}\underline{a}'M\underline{a}\right)$; conversely, if $\underline{v}'M\underline{v} \sim \chi^2(s, \lambda)$ for some λ, then M is idempotent of rank s.

Proof: Following the hint, we have two goals: (1) $\Gamma A\Gamma'$ is idempotent of rank s, and (2) $\underline{x}'A\underline{x} \overset{d}{=} \underline{v}'\Gamma A\Gamma'\underline{v}$ for some $\underline{v} \sim N(\underline{a}, I_k)$ with $\underline{a}'\Gamma A\Gamma'\underline{a} = \underline{\mu}'A\underline{\mu}$.

(1) $\Gamma A\Gamma'$ is idempotent of rank s: Since $A\Sigma$ is idempotent,

$$(A\Sigma A)\Gamma'\Gamma = A\Sigma A\Sigma = A\Sigma = A\Gamma'\Gamma,$$

so Lemma 1.3(i), with Γ in the role of X, gives $A\Sigma A\Gamma' = A\Gamma'$. Premultiplying by Γ,

$$(\Gamma A\Gamma')^2 = \Gamma A(\Gamma'\Gamma)A\Gamma' = \Gamma(A\Sigma A\Gamma') = \Gamma A\Gamma',$$

so the symmetric matrix $\Gamma A\Gamma'$ is idempotent. By the Corollary to Lemma 2.5 (twice) and the cyclic property of the trace,

$$\text{rank}(\Gamma A\Gamma') = \text{tr}(\Gamma A\Gamma') = \text{tr}(A\Gamma'\Gamma) = \text{tr}(A\Sigma) = \text{rank}(A\Sigma) = s.$$

(2) The representation: Since $\underline{\mu} \in \mathcal{C}(\Sigma)$, $\underline{\mu} = \Sigma\underline{d}$ for some $\underline{d} \in \mathbb{R}^r$, so $\underline{\mu} = \Gamma'\Gamma\underline{d} = \Gamma'\underline{a}$ with $\underline{a} \equiv \Gamma\underline{d} \in \mathbb{R}^k$. By the definition, $\underline{x} \overset{d}{=} \underline{\mu} + \Gamma'\underline{z} = \Gamma'(\underline{z} + \underline{a})$, where $\underline{z} \sim N(\underline{0}_k, I_k)$. Writing $\underline{v} \equiv \underline{z} + \underline{a} \sim N(\underline{a}, I_k)$ (Theorem 3.5), it then follows that

$$\underline{x}'A\underline{x} \overset{d}{=} (\Gamma'\underline{v})'A(\Gamma'\underline{v}) = \underline{v}'(\Gamma A\Gamma')\underline{v} \quad \text{and} \quad \underline{a}'\Gamma A\Gamma'\underline{a} = (\Gamma'\underline{a})'A(\Gamma'\underline{a}) = \underline{\mu}'A\underline{\mu}.$$

Combining (1)–(2), Theorem 3.16 (case $\Sigma = I$), applied to $\underline{v}$ and $\Gamma A\Gamma'$, gives

$$\underline{x}'A\underline{x} \overset{d}{=} \underline{v}'(\Gamma A\Gamma')\underline{v} \sim \chi^2\left(s, \frac{1}{2}\underline{a}'\Gamma A\Gamma'\underline{a}\right) = \chi^2\left(s, \frac{1}{2}\underline{\mu}'A\underline{\mu}\right).$$

Said differently, $\underline{\mu} \in \mathcal{C}(\Sigma)$ lets the mean be absorbed into the standard normal vector, which is what carries the sufficiency half of Theorem 3.16 over to singular Σ.

Q6

Let $\underline{x} \sim N(\underline{0}, I)$, and take A and B to be two symmetric matrices such that $\underline{x}'A\underline{x} \sim \chi^2(p)$ and $\underline{x}'B\underline{x} \sim \chi^2(q)$, where $p > q$. Show, by completing the following three steps, that $\underline{x}'(A-B)\underline{x} \sim \chi^2(p-q)$ if and only if the correlation $\text{corr}(\underline{x}'A\underline{x}, \underline{x}'B\underline{x}) = q/\sqrt{pq}$.

Note: Let n denote the length of $\underline{x}$, so $\underline{x} \sim N(\underline{0}_n, I_n)$ and A, B are $n \times n$. Since $\underline{x}'A\underline{x} \sim \chi^2(p)$ and $\underline{x}'B\underline{x} \sim \chi^2(q)$, the converse in Theorem 3.16 makes A and B idempotent of ranks p and q, resp. Hence $\text{tr}(A) = p$ and $\text{tr}(B) = q$ (Corollary to Lemma 2.5), $I_n - A$ is symmetric and idempotent as well, and $p - q \geq 1$ (as p and q are ranks).

We utilize the following results for (a)–(c):

- (Q2): for symmetric C and D, $\text{cov}(\underline{x}'C\underline{x}, \underline{x}'D\underline{x}) = 2\,\text{tr}(C\Sigma D\Sigma) + 4\underline{\mu}'C\Sigma D\underline{\mu}$, which here ($\underline{\mu} = \underline{0}_n$, $\Sigma = I_n$) reads $2\,\text{tr}(CD)$.
- Theorem 3.16 (case $\Sigma = I$; stated in Q5) and the Corollary to Lemma 2.5 (stated in Q4).
- Moments of the χ^2 (Section 3.3): if $u \sim \chi^2(m, \lambda)$, then $\text{Var}(u) = 2(m + 4\lambda)$.
- Lemma 1.2: $\text{tr}(X'X) = 0$ implies $X = 0$.
- Lemma 3.6: if F, G, and $F + G$ are all idempotent, then $FG = 0$.

(a)

Show that $\text{tr}[(I - A)B] = 0$ if and only if $\text{corr}(\underline{x}'A\underline{x}, \underline{x}'B\underline{x}) = q/\sqrt{pq}$. [Hint: Consider $\text{cov}(\underline{x}'(A-B)\underline{x} + \underline{x}'B\underline{x}, \underline{x}'B\underline{x})$]

Answer

Proof: By (Q2) and $B^2 = B$,

$$\text{cov}(\underline{x}'(A-B)\underline{x}, \underline{x}'B\underline{x}) = 2\,\text{tr}[(A-B)B] = 2\,\text{tr}(AB - B) = -2\,\text{tr}[(I_n - A)B],$$

while $\text{Var}(\underline{x}'A\underline{x}) = 2p$ and $\text{Var}(\underline{x}'B\underline{x}) = 2q$ (Section 3.3 with $\lambda = 0$). Following the hint, bilinearity of the covariance gives

$$\begin{aligned}\text{cov}(\underline{x}'A\underline{x}, \underline{x}'B\underline{x}) &= \text{cov}(\underline{x}'(A-B)\underline{x} + \underline{x}'B\underline{x}, \underline{x}'B\underline{x}) \\ &= \text{cov}(\underline{x}'(A-B)\underline{x}, \underline{x}'B\underline{x}) + \text{Var}(\underline{x}'B\underline{x}) = 2q - 2\,\text{tr}[(I_n - A)B],\end{aligned}$$

so that

$$\text{corr}(\underline{x}'A\underline{x}, \underline{x}'B\underline{x}) = \frac{2q - 2\,\text{tr}[(I_n - A)B]}{\sqrt{2p \cdot 2q}} = \frac{q - \text{tr}[(I_n - A)B]}{\sqrt{pq}}.$$

Hence $\text{corr}(\underline{x}'A\underline{x}, \underline{x}'B\underline{x}) = q/\sqrt{pq}$ if and only if $\text{tr}[(I_n - A)B] = 0$.

(b)

Show that $(I - A)B = 0$ if and only if $\text{tr}[(I - A)B] = 0$.

Answer

Proof: ($\Rightarrow$) If $(I_n - A)B = 0$, then $\text{tr}[(I_n - A)B] = \text{tr}(0) = 0$.

($\Leftarrow$) Write $M \equiv B(I_n - A)$, so that $M' = (I_n - A)B$ by symmetry. Since $(I_n - A)^2 = I_n - A$ and $B^2 = B$, the cyclic property of the trace gives

$$\text{tr}[(I_n - A)B] = \text{tr}[(I_n - A)(I_n - A)BB] = \text{tr}[(I_n - A)BB(I_n - A)] = \text{tr}(M'M).$$

So if $\text{tr}[(I_n - A)B] = 0$, then $\text{tr}(M'M) = 0$, and Lemma 1.2 gives $M = 0$; hence $(I_n - A)B = M' = 0$.

Thus $(I_n - A)B = 0$ if and only if $\text{tr}[(I_n - A)B] = 0$.

(c)

Show that $\underline{x}'(A-B)\underline{x} \sim \chi^2(p-q)$ if and only if $(I-A)B = 0$.

Answer

Proof: Since $A-B$ is symmetric, $\underline{x} \sim N(\underline{0}_n, I_n)$, and $p-q \geq 1$, Theorem 3.16 (case $\Sigma = I$) gives $\underline{x}'(A-B)\underline{x} \sim \chi^2(p-q)$ if and only if $A-B$ is idempotent of rank $p-q$ (the noncentrality $\frac{1}{2}\underline{0}_n'(A-B)\underline{0}_n$ is zero). It then suffices to show that $A-B$ is idempotent of rank $p-q$ if and only if $(I_n - A)B = 0$.

($\Rightarrow$) If $A-B$ is idempotent, then $A-B$, B, and $(A-B)+B = A$ are all idempotent, so Lemma 3.6 gives $(A-B)B = 0$, i.e., $AB - B = 0$, i.e., $(I_n - A)B = 0$.

($\Leftarrow$) If $(I_n - A)B = 0$, then $AB = B$, and transposing (by symmetry) $BA = B$. Hence

$$(A-B)^2 = A^2 - AB - BA + B^2 = A - B - B + B = A - B,$$

so $A-B$ is idempotent, with $\text{rank}(A-B) = \text{tr}(A-B) = p-q$ (Corollary to Lemma 2.5).

Combining (a)–(c),

$$\underline{x}'(A-B)\underline{x} \sim \chi^2(p-q) \overset{\text{(c)}}{\iff} (I_n - A)B = 0 \overset{\text{(b)}}{\iff} \text{tr}[(I_n - A)B] = 0 \overset{\text{(a)}}{\iff} \text{corr}(\underline{x}'A\underline{x}, \underline{x}'B\underline{x}) = \frac{q}{\sqrt{pq}},$$

as claimed.